ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY
A.S Nagar, Elupatti– Thanjavur.
(Approved by AICTE-New Delhi, Affiliated to Anna University, Chennai.)

Department of Mechanical Engineering

CONTINUOUS INTERNAL ASSESSMENT TEST-II, OCT - 2026

Reg.No	8	2	1	9									
Regulation	**R2025**												
Course Code-Name	**ME25C03 & INTRODUCTION TO MECHANICAL ENGINEERING**									**SET- A**			
Year / Semester	**I & I**									**Total Marks:100**			
Date - Time	**23. 10. 2026 / 9.30 to 12.30**												
Course Instructor	**Mr. B.Nagendran AP/MECH**												

C201.1 – Explain core mechanical engineering concepts.

C201.2 – Apply basic engineering calculations in mechanical engineering.

L1-Remember , L2-Understand , L3-Apply , L4-Analyze , L5 -Evaluate , L6-Create

Answer All Questions
PART – A (10 × 1 = 10 Marks & 10 × 3 = 30 Marks)

Q.NO	QUESTIONS	Marks	CO	LEVEL
1.	Which of the following is defined as a push or pull that can change an object's state of rest or motion? a) Motion b) Energy c) Force d) Power	**1**	C201.1	**L1**
2.	What is the mathematical formula for force? a) F = m/a b) F = m × v c) F = m × a d) F = W/t	**1**	C201.1	**L1**
3.	What is the SI unit of force? a) Dyne b) Newton c) Joule d) Watt	**1**	C201.1	**L1**
4.	What is the SI unit of energy? a) Newton b) Joule c) Watt d) Pascal	**1**	C201.1	**L2**
5.	What is the formula for potential energy? a) ½mv² b) mgh c) F × d d) W/t	**1**	C201.1	**L2**
6.	In SI units, the unit of stress is: a) Newton b) Joule c) Pascal d) Watt	**1**	C201.2	**L2**
7.	What is the mathematical formula for stress? a) $\sigma = P \times A$ b) $\sigma = A/P$ c) $\sigma = P/A$ d) $\sigma = P^2A$	**1**	C201.2	**L2**
8.	The ratio of change in volume to the original volume is called: a) Linear strain b) Lateral strain c) Volumetric strain d) Shear strain	**1**	C201.2	**L1**
9.	Poisson's ratio is defined as the ratio of: a) Linear strain to lateral strain b) Lateral strain to linear strain c) Volumetric strain to linear strain d) Shear strain to linear strain	**1**	C201.2	**L2**
10.	Poisson's ratio of a material can never be greater than: a) 0.3 b) 0.4 c) 0.5 d) 1.0	**1**	C201.2	**L1**
11.	Define force. State its SI unit and explain its types.	**3**	C201.1	**L2**
12.	Define motion and state how it is described.	**3**	C201.1	**L2**
13.	Define energy and list its various types.	**3**	C201.1	**L1**

14.	Define work and state its SI unit.	**3**	C201.1	**L2**
15.	Define power and state its SI unit.	**3**	C201.1	**L1**
16.	Define stress and state its mathematical formula.	**3**	C201.2	**L1**
17.	What is strain? How is it calculated?	**3**	C201.2	**L1**
18.	Define Young's modulus.	**3**	C201.2	**L2**
19.	Define Poisson's ratio and state its importance.	**3**	C201.2	**L2**
20.	Explain shear stress with an example.	**3**	C201.2	**L1**
PART – B (5 × 12 = 60 Marks)				
21.a	Explain the history and evolution of mechanical engineering.	**12**	C201.1	**L3**
(OR)				
21.b	Explain the following: (i) Force (ii) Motion (iii) Energy (iv) Work.	**12**	C201.1	**L3**
22.a	Discuss the various dimensions and units used in mechanical engineering.	**12**	C201.1	**L3**
(OR)				
22.b	Explain in detail the ethics and professionalism in engineering.	**12**	C201.1	**L3**
23.a	Discuss in detail the SI system of units.	**12**	C201.1	**L3**
(OR)				
23.b	Explain the following: (i) Stress (ii) Strain (iii) Poisson's ratio.	**12**	C201.1	**L3**
24.a	Discuss in detail the various types of stresses.	**12**	C201.2	**L4**
(OR)				
24.b	Define elasticity and plasticity. Explain them with a stress–strain diagram.	**12**	C201.2	**L4**
25.a	Define and explain the mechanical properties of materials.	**12**	C201.2	**L4**
(OR)				
25.b	Explain the method of determining the forces (stresses) in the members of a truss with a suitable example.	**12**	C201.2	**L4**

Course Instructor **HOD** **IQAC Coordinator**